

Analysis of Washington Solid Waste and Recycling Data

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Abstract

Washington State is facing a rapid increase in waste generation, as well as inadequate funding and regulatory support as the population grows. Waste management has been an important topic, highlighting the effort to reduce solid waste, mitigate environmental damage, and preserve ecosystems. This study analyzes time-series trends in solid waste generation and recycling rates across Washington State counties. The study emphasizes real-world data acquisition, preparation, and analysis supported by visualizations. Our goal is to identify whether urban counties generate more or less waste per capita than rural ones and examine how that gap has evolved over time. These findings aim to inform county-level waste management planning and policy decisions.

Keywords: data science, data analysis, visualization, waste, recycling

1. INTRODUCTION

Washington State produces millions of tons of solid waste annually which ties to the growth of the population, economic activity and the regulatory environment that governs how counties collect, recycle and dispose of materials. As the population continues to grow, county waste management agencies face pressure to plan infrastructure, recycling programs and disposal capacity under conditions of uneven funding and inconsistent reporting practices. Understanding how waste generation differs from densely populated urban counties and lower density rural counties is essential for allocating resources, setting recycling targets and informing state level policy.

This study analyzes county level waste data from the Washington State Department of Ecology in combination with U.S. Census Bureau population

estimates, with the objective of identifying whether urban counties generate more waste per capita than rural counties and how this has evolved from 2015 through 2018.

2. LITERATURE REVIEW

Data acquisition and data analysis are ongoing topics that are heavily studied because of their impact on data-related insights and decision-making.

The study by Malik (2023) explores time series analysis and forecasting techniques with the goal of finding hidden insights and anticipating future outcomes. The author emphasizes the importance of interpretability and how transparency through analysis is important in critical decision-making. Techniques to improve interpretability include feature importance analysis, local explanations, visualizations, rule-

based models, and sensitivity analysis. Interpretability also aligns with ethical principles with accountability in mind, though a balance needs to be struck as complex models can have better performance.

In the study by Manonmani (2025), the author discusses data acquisition fundamentals and methodology. The importance of data collection is growing as organizations are increasingly relying on data-driven insights. Important aspects of data acquisition include data quality, scalability, security, timeliness, and data integration. Gathering data from multiple sources can increase complexity. Poorly extracted data can lead to data quality issues such as incomplete or inaccurate data, leading to issues such as poor machine learning model performance or inaccurate insights.

In *Waste in Washington* by the Washington Department of Ecology (2025), the authors analyze the state of the Solid Waste Management program. They analyze the environmental and financial impacts of waste, discuss waste reduction, and educate about the state plans to make changes in waste policies. Data analysis surrounding waste management has been used to inform state plans. In this case, the data helped the state move towards sustainable material management and focus on environmental impact and sustainability.

In the study by Vines et al (2023), the authors examine three different modeling approaches used to estimate national municipal solid waste (MSW) flows in the U.S., industry-based material flow analysis, waste-extended input-output modeling and aggregated regional waste reporting. They evaluate each approach against five criteria, data quality, flow totality, update frequency, sensitivity to disruption and granularity and conclude that no single approach satisfies all policy needs.

3. METHODOLOGY

This study addresses three technical contributions: parsing and harmonizing raw waste data published as multi-sheet Excel workbooks whose schemas shift from year to year; integrating the cleaned waste data with external population and population density data using county name as a stable join key; and applying a classification scheme that

distinguishes urban, suburban, and rural counties based on population density thresholds to support exploratory analysis and visualization.

Data Acquisition

The three datasets acquired in this study are:

- The county population estimate data (PEPANNRES) collected from the U.S. Census Bureau. We focus on the years 2014–2018, providing a five-year population growth trend.
- The annual Washington Solid Waste report from the Washington State Department of Ecology. Consistent with the Census data, we focus on the years 2014–2018.
- The county population density data collected from the Washington State Office of Financial Management. Since an exact density figure for 2014–2018 is not available, we use the 2020 estimate as the closest available approximation for classifying counties by urbanization level.

By merging the three datasets on county name as the join key, the resulting DataFrame captures the total population, total solid waste disposal in tons, and the urbanization level for each county across each year.

```
# Merge all five waste datasets together on County
df_merged = data2014_m \
    .merge(data2015_m, on='County', how='outer') \
    .merge(data2016_m, on='County', how='outer') \
    .merge(data2017_m, on='County', how='outer') \
    .merge(data2018_m, on='County', how='outer')

# Merge with population and density data
df_final = df_merged.merge(df_pop, on='County', how='left') \
    .merge(df_dense, on='County', how='inner')

# Reorder columns
df_final = df_final[['County', 'Total_2014', 'Population_2014',
                    'Total_2015', 'Population_2015',
                    'Total_2016', 'Population_2016',
                    'Total_2017', 'Population_2017',
                    'Total_2018', 'Population_2018',
                    'Pop_Den', 'Urban_Level']]
```

Figure 1: DataFrame Integration — Merging Solid Waste, Population, and Urbanization Data by County

Data Preparation and Cleaning

The population dataset is reduced to the years 2014–2018 by dropping irrelevant columns. Rows containing Oregon county data are also removed. In the County column, all county names are stripped from the format "[Name] County, Washington" to "[Name]" to align with the solid

waste dataset for merging. Population values are converted from string format to integers by removing the thousand-separator comma and applying .astype(int).

In the solid waste dataset, each sheet contains annual data from 2010 to 2018. The data is presented as a pivot table with a two-level header structure. The columns represent the types of solid waste, and the rows represent county names, each followed by its associated waste management facilities as sub-rows. For this study, we extract the Grand Total column and the county-level rows. Since county rows are not uniformly spaced due to varying numbers of facilities per county, we extract county-level rows by cross-referencing a list of county names derived from the population dataset, filtering the solid waste dataset to retain only rows whose names match.

```
# Match county names from df_waste_2015 to df_pop, return a new DF
data2015 = df_waste_2015[df_waste_2015['County'].isin(counties)]
data2015.head(10)
```

	County	Appliances/White Goods	Asbestos	Ash (other than special incinerator ash)	Asphalt/Concrete
0	Adams	NaN	16.35	NaN	NaN
5	Asotin	NaN	1.35	NaN	NaN
9	Benton	NaN	26.22	NaN	NaN
18	Chelan	NaN	689.13	NaN	NaN
26	Clallam	NaN	11.62	10121.70	NaN
30	Clark	NaN	5.79	1659.17	NaN
38	Columbia	NaN	0.38	NaN	NaN
41	Cowlitz	NaN	5.66	87846.00	NaN
45	Douglas	NaN	124.96	NaN	NaN
53	Ferry	NaN	5.10	NaN	NaN

Figure 2: County-Level Solid Waste DataFrame After Filtering by County Name

The population density dataset is cleaned and reduced to two columns: the county name, renamed to County, and the 2020 population density estimate, renamed to Pop_Den. Using pd.cut(), counties are categorized into three urbanization levels based on population density thresholds: Rural (0–100 people per square mile), Suburban (100–500 people per square mile), and Urban (500+ people per square mile). These thresholds are defined to reflect the density distribution across Washington's 39 counties, where only the most densely populated counties qualify as Urban. The resulting urbanization classifications are stored in a new Urban_Level column.

```
# Add a column categorizing Urbanization level
#
df_dense['Urban_Level'] = pd.cut(x=df_dense['Pop_Den'],
                                bins=[0,100,500, float('inf')],
                                labels=['Rural', 'Suburban', 'Urban'])
# Ensure County Names format aligns
df_dense['County'] = df_dense['County'].str.strip()
df_dense.head()
```

	County	Pop_Den	Urban_Level
1	Adams	10.71	Rural
2	Asotin	35.04	Rural
3	Benton	121.68	Suburban
4	Chelan	27.09	Rural
5	Clallam	44.37	Rural

Figure 3: Urbanization Level Classification Using pd.cut() Based on Population Density Thresholds

The density dataset is then merged with the final DataFrame containing county-level solid waste totals and population estimates, producing a unified dataset ready for exploratory analysis

Exploratory Analysis and Visualization

[to be continued]

Per Capita Waste by County

- Which counties generate the most and least waste per person? Bar charts and ranked comparisons

Urbanization vs Per Capita Waste

- Correlation analysis — population density vs per capita disposal Group comparison — urban vs rural county clusters Scatter plots and box plots by urbanization group

Trends Over Time (2014–2018)

- How has per capita waste changed over the five years? Time-series line plots by county and urbanization group

Key Findings from Exploration

- Summary of patterns, outliers, and notable observations

4. DISCUSSION

5. CONCLUSION

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